

3F7 Lab: CamZIP

A data compression lab

Lab contents

Lab:

- Data structures to hold codes and trees and how to visualise them
- Shannon-Fano code
- Huffman algorithm
- Arithmetic coding

FTR:

- Adaptive / universal compression
- Conditional (context-based) compression

Lab format

- Write / complete software to implement data compression algorithms
- Try out and use tools we prepared for you (e.g. tree to code conversion)
- Code templates are provided for all steps. You need to complete them
- Lab software templates are in Python
- You can write code in any language you prefer

Lab organisation

- All files on Moodle and need to be downloaded and unpacked / uncompressed (using gzip, not CamZIP 😊)
- Lab can be done in your own time on your own computer!
- Online helpdesk sessions Wednesdays 11 am, details on moodle. Please note that the helpdesk closes if unattended for 10 minutes, so try to make it as close to 11 as possible if you want to get help

Assessment

- Lab is assessed based on a multiple choice quiz
- Recommended time to do the lab is shortly after you learn about arithmetic coding in lectures
- Immediate feedback is provided on moodle, as well as an indicative percentual mark
- Marks are mapped to the 5 point scale after moderation once all have completed their lab. The average mark in 2023 was 4.2/5
- Deadline: **Wednesday of W6, 4 pm**
- Please provide general feedback about this lab via the quiz or via the anonymous feedback facility. YOUR FEEDBACK IS HIGHLY APPRECIATED particularly when constructive! I almost always implement any suggestions received via feedback.

FTR

- Investigate the theory of data compression for unknown sources with memory (“universal contextual compression”)
- Marking based on a report
- Deadline: FTR deadline as published by department (usually Wednesday of W9, 4 pm but can vary, please check department info)